

CIT305: Social Media Applications

Section A

1. What is a chatbot?

A chatbot is a software application that simulates human conversation through text or voice, using rule-based logic or AI (such as natural language processing) to understand user input and generate relevant responses. Businesses use chatbots to automate tasks like answering questions, booking services, or providing support.

2. What is meant by a conversational interface?

A conversational interface is a user interface that allows people to interact with software using natural, everyday language (typed or spoken), rather than clicking through menus, forms, or buttons. It mimics the flow of a human conversation, e.g. chatbots and voice assistants.

3. Identify the primary problem that Sephora's chatbot was designed to solve.

Sephora wanted to make it easier and more convenient for customers to book in-store makeover appointments, since doing this by phone or website was slow and inconvenient. The chatbot gave customers a fast, familiar channel (Facebook Messenger) to book appointments and get beauty guidance.

4. How did Sephora's chatbot help customers book makeovers?

Through Facebook Messenger, the chatbot let customers pick a nearby Sephora store, view available appointment time slots, and confirm a makeover booking all within the chat, without needing to call the store or navigate a website.

5. What type of customer services could Sephora's chatbot provide besides appointment booking?

It could help customers locate nearby stores, check product and service availability, and offer personalised beauty tips and product recommendations based on customer preferences.

6. What was the main operational challenge faced by KLM that led to the development of BlueBot?

KLM was receiving a very large and fast-growing volume of customer service messages across digital and social channels (such as Twitter and Facebook), especially during flight disruptions. Human agents alone could not respond quickly enough to this volume, so KLM needed an automated solution.

7. List two tasks that KLM's BlueBot could perform.

- 1) Send booking confirmations and flight-related information (such as check-in details and boarding passes).
- 2) Answer frequently asked questions and provide real-time flight status updates.

8. What is meant by a chatbot handover feature?

A handover feature is the ability of a chatbot to transfer a conversation to a human customer service agent when a query is too complex, sensitive, or outside the bot's capability, while passing along the conversation history so the customer does not have to repeat themselves.

9. State two benefits of chatbots to businesses.

- 1) 24/7 availability, allowing customers to get help at any time without waiting for business hours.
- 2) Lower operational costs and greater scalability, since a single chatbot can handle many customer conversations at once, reducing the workload on human staff.

10. Give two differences between Sephora's chatbot and KLM's BlueBot.

- 1) Purpose: Sephora's chatbot focuses on retail/beauty services (appointment booking and beauty advice), while BlueBot focuses on airline customer service (flight updates and travel-related inquiries).
- 2) Scale of use: BlueBot was built mainly to manage a very high volume of urgent, time-sensitive travel inquiries, whereas Sephora's chatbot handles a smaller, more personal, advice-driven interaction focused on customer engagement and shopping.

Section B

11. Sephora Chatbot

a) Explain the problem faced by Sephora before introducing its chatbot.

Before the chatbot, customers who wanted a makeover appointment had to call the store or use the website, which was time-consuming and not always convenient. Sephora needed a faster, more accessible way for customers — many of whom were already active on social media to book services and get beauty advice.

b) Explain how the chatbot improved the customer experience.

- Made booking simple and fast by letting customers choose a store and time slot directly inside Facebook Messenger.
- Provided 24/7 access to beauty tips and product recommendations, personalising the shopping experience.
- Reduced friction between browsing and buying, increasing engagement and encouraging repeat visits, which strengthened brand loyalty.

12. KLM BlueBot

a) Explain why KLM needed to develop BlueBot.

KLM faced a surge in customer inquiries through digital and social channels, particularly during flight disruptions, and its human agents could not keep pace with the volume and speed of messages. BlueBot was developed to automate responses and manage this scale efficiently.

b) Explain three benefits KLM gained from using BlueBot.

- Faster response times, since BlueBot could instantly deliver booking confirmations and flight updates.
- Reduced workload on human agents, freeing them to focus on complex cases via the handover feature.
- Improved customer satisfaction and cost efficiency, as BlueBot could handle many conversations simultaneously at lower cost than scaling up human staff.

13. Compare the Two Chatbots

Area	Sephora Chatbot	KLM BlueBot
Main purpose	Assist retail/beauty customers with bookings and product advice	Manage high-volume airline customer service inquiries
Customer needs	Convenient makeover booking, beauty tips, store info	Flight updates, booking confirmations, quick answers to travel questions
Services provided	Appointment booking, store locator, personalised beauty advice	Boarding passes, flight status, baggage info, FAQs, booking confirmations
Business benefits	Increased in-store visits, stronger customer engagement, brand loyalty	Reduced agent workload, faster response times, cost savings, scalability

Human-agent involvement	Lower complexity; human help mainly in-store	Handover feature routes complex queries to live agents
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14. Chatbot Challenges

Discuss five potential challenges or limitations a business may face when implementing a chatbot:

- Limited understanding of complex or ambiguous queries chatbots can misinterpret unusual phrasing, slang, or multi-part questions.
- Lack of human empathy chatbots may struggle to handle emotionally sensitive situations (e.g. complaints, bereavement travel) as well as a human agent.
- High development and maintenance costs — building, training, and continually updating a chatbot requires ongoing investment.
- Integration difficulties connecting the chatbot with existing databases, booking systems, or CRM platforms can be technically challenging.
- Risk of customer frustration if the bot fails to resolve an issue or loops without a clear handover to a human, customers may become dissatisfied and lose trust in the brand.

15. Chatbots and Customer Service

Explain how chatbots can improve customer service and operational efficiency, using Sephora and KLM as examples.

Chatbots improve customer service by providing instant, round-the-clock responses, reducing waiting times and freeing human staff to focus on complex or sensitive issues. They also improve operational efficiency by automating repetitive tasks, handling many conversations simultaneously, and lowering the cost per customer interaction.

Sephora's chatbot illustrates this by letting customers book makeovers and get beauty advice instantly through Messenger, improving convenience and driving store engagement without requiring extra staff. KLM's BlueBot illustrates this at larger scale it manages a high volume of urgent travel-related inquiries automatically, using a handover feature to escalate complex cases to human agents, which keeps response times fast while still ensuring quality support for difficult queries.

Section C

16. Case Study – Sephora

a) Analyze how Sephora's chatbot addressed the organization's operational problem.

Sephora's core problem was making its in-store services especially makeover bookings easily accessible to a customer base that was increasingly shopping and communicating through mobile and social platforms. By deploying a chatbot on Facebook Messenger, a channel customers already used daily, Sephora removed the friction of phone calls or navigating a separate booking website. The chatbot automated the appointment process (store selection, time-slot checking, and confirmation), which reduced manual work for staff and shortened the path from customer interest to an actual store visit, directly solving the accessibility and convenience problem.

b) Discuss how the chatbot contributed to customer engagement and brand loyalty.

Because the chatbot offered personalised beauty advice and quick, useful interactions, customers engaged with the Sephora brand more frequently and in a more conversational, less transactional way. This ongoing,

low-effort engagement available anytime helped build a stronger relationship with the brand and encouraged repeat bookings and store visits, reinforcing loyalty.

c) Identify and explain two possible limitations of this chatbot.

- Limited platform reach: customers who don't use Facebook Messenger may not be able to access the service, excluding part of the customer base.
- Limited personalisation depth: for complex beauty needs (e.g. skin conditions, allergies), a chatbot's advice may be too generic, and customers may still need to consult an in-store beauty advisor.

17. Case Study – KLM BlueBot

a) Explain the operational problems KLM faced before implementing BlueBot.

KLM was receiving an overwhelming number of customer messages across social and digital channels, particularly during flight delays or cancellations, when demand for real-time information spikes sharply. Human agents alone could not respond to this volume quickly enough, leading to slower service and potential customer dissatisfaction.

b) Discuss how BlueBot improved KLM's efficiency and customer satisfaction.

BlueBot automated routine, high-volume tasks such as sending booking confirmations, boarding passes, and flight status updates, allowing thousands of customers to be served simultaneously without waiting for a human agent. This significantly cut response times and operating costs. Customer satisfaction improved because travellers received instant, accurate information exactly when they needed it (e.g. during disruptions), and the handover feature ensured that anyone with a more complex problem could still reach a human agent without losing context.

c) Explain why the human-agent handover feature was important to the success of BlueBot.

Not every customer issue can be resolved by automated responses – some require judgement, empathy, or the authority to make exceptions (e.g. rebooking after a missed connection). The handover feature let BlueBot recognise these situations and pass the conversation, along with its history, to a human agent. This balance of automation and human support meant KLM could scale its service without sacrificing quality for customers with genuinely complex needs, which was essential to maintaining trust in the system.

18. Critical Comparison

"Sephora's chatbot and KLM's BlueBot demonstrate how organizations can use conversational interfaces for different strategic purposes." Discuss this statement by comparing: Purpose and scope, Customer experience, Automation, Business efficiency, Personalization, Human support.

Purpose and scope: Sephora's chatbot was built around a retail, relationship-driven purpose helping customers book services and receive beauty guidance to drive store visits and sales. BlueBot's purpose was operational and service-driven – managing an airline's very high volume of time-critical travel inquiries efficiently across digital channels.

Customer experience: Sephora's bot creates a personal, advisory experience, similar to speaking with a beauty consultant, while BlueBot creates a fast, transactional experience focused on delivering accurate travel information quickly, especially valuable during stressful situations like delays.

Automation: Both automate repetitive tasks, but BlueBot operates at a much larger scale given the volume and urgency of airline customer queries, whereas Sephora's chatbot handles a narrower, more curated set of interactions around bookings and advice.

Business efficiency: BlueBot delivers efficiency mainly by cutting response times and staffing costs during high-demand periods (e.g. disruptions), while Sephora's chatbot delivers efficiency by streamlining the booking funnel and reducing manual scheduling work for store staff.

Personalization: Sephora leans more heavily on personalisation tailoring beauty advice to individual customers as a core selling point, while BlueBot's personalisation is more limited, mainly tied to a customer's specific booking or flight.

Human support: Both use human agents as a backstop, but the handover feature is more central to BlueBot's design, given the higher stakes and complexity of travel disruptions, whereas Sephora's chatbot relies more on in-store staff for hands-on, higher-touch beauty consultations.

In conclusion, both cases show that conversational interfaces are not one-size-fits-all, the same underlying chatbot technology can be shaped to serve very different strategic goals ,relationship-building and sales for Sephora, and large-scale operational efficiency for KLM ,depending on the needs of the business and its customers.